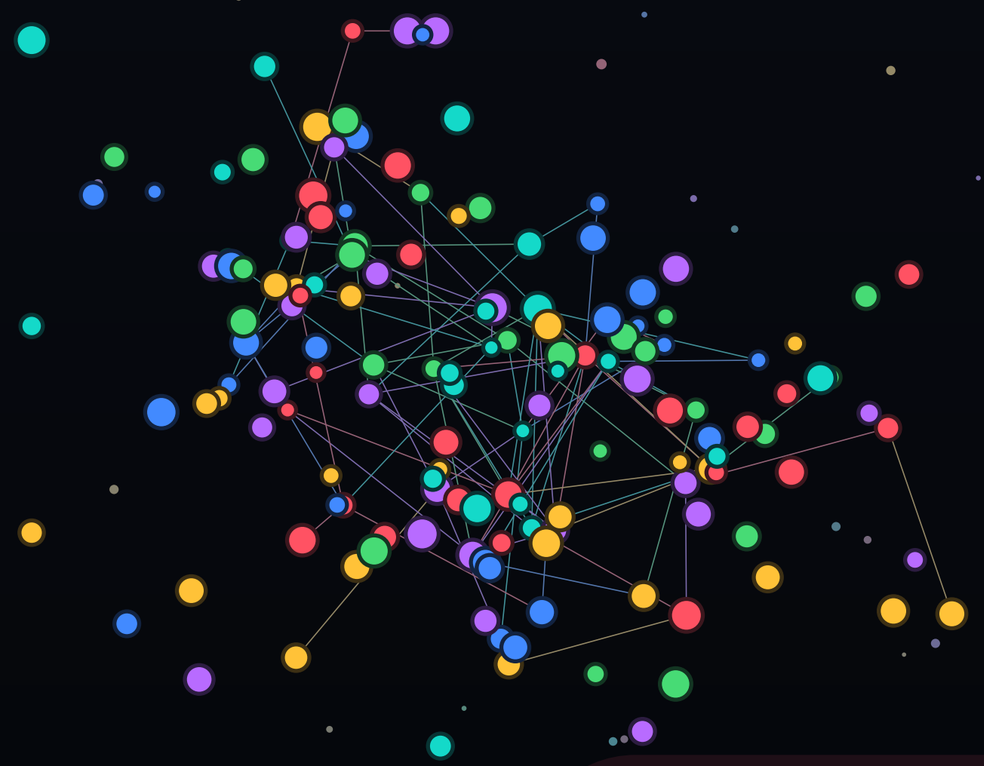
A FIRST LOOK

Particle Life: tiny rules, living motion

A particle life system is not scripted animation. It is a crowd of simple dots that repeatedly pull, push, and organize themselves.



thousands of local interactions

THE INGREDIENTS

Nothing starts out intelligent

Each particle only carries a few numbers: where it is, how fast it is moving, and what species or color it belongs to.

- Position says where the particle is.
- Velocity says where it is heading next.
- Species decides which rules affect it.

position
where am I?

velocity
where next?

species
which rules?

THE RULEBOOK

The whole personality is a table of relationships

For every pair of species, one number says attract, repel, or ignore. That table is enough to create surprisingly rich behavior.

- Positive values pull particles together.
- Negative values push them apart.
- The relationship can be directional.

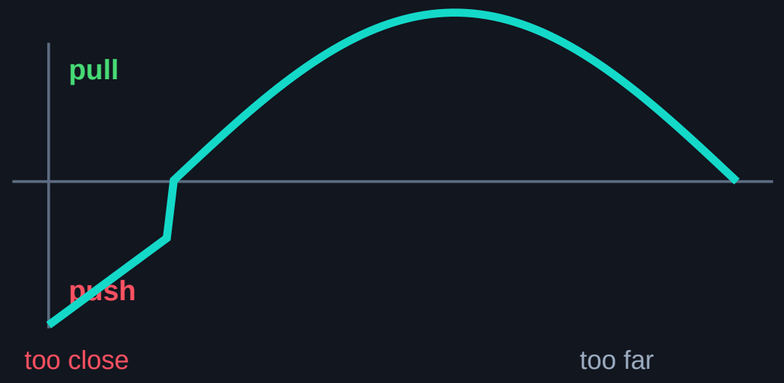
THE MOTION

Every frame repeats the same small question

For each nearby particle: how far away is it, what species is it, and should I move toward it or away from it?

One pairwise force

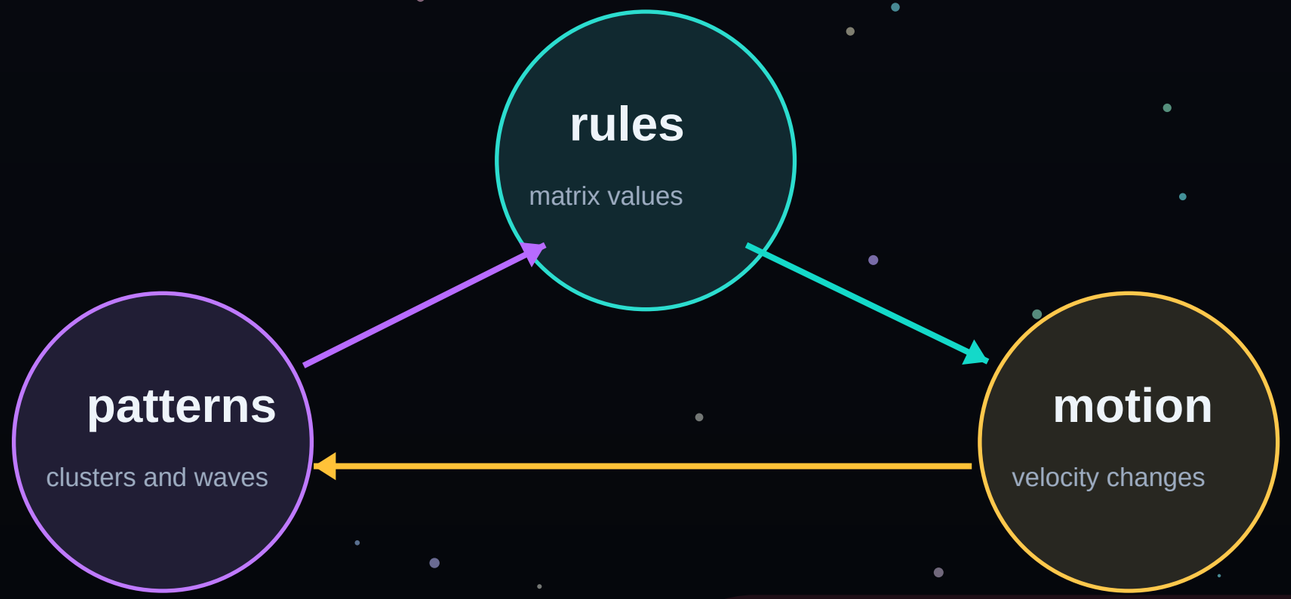
Distance changes how strongly particles react



THE RESULT

Patterns appear without being drawn

Clusters, rings, streams, and waves are side effects of the rules feeding back into motion.



TAKEAWAY

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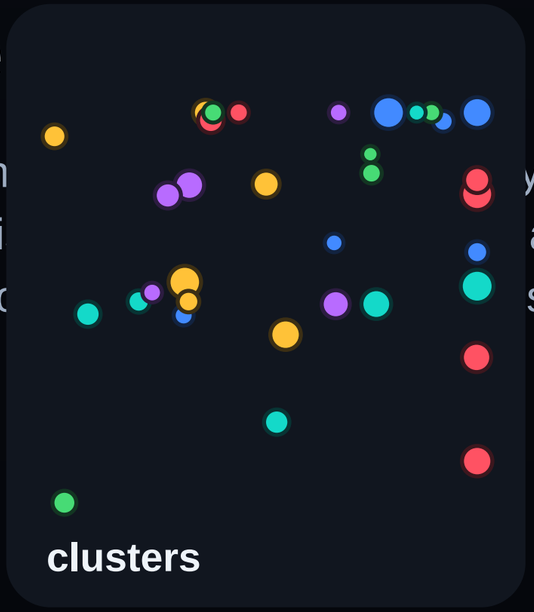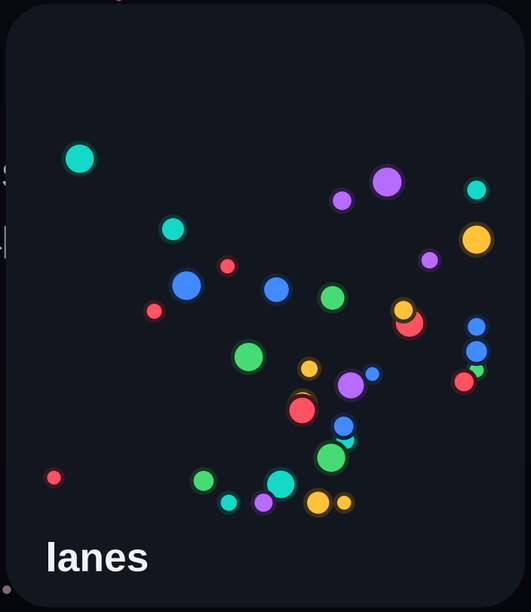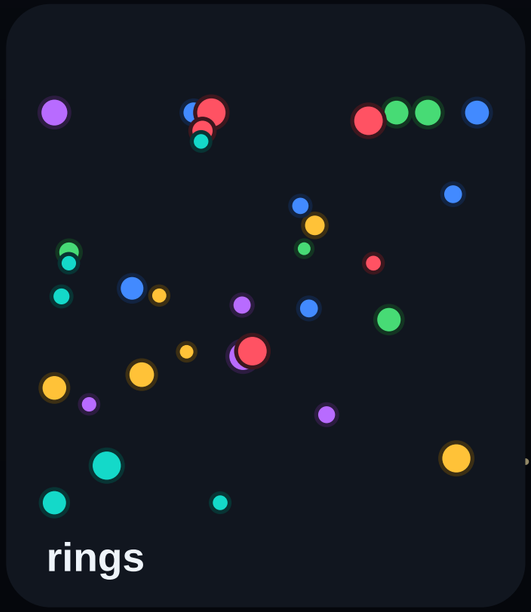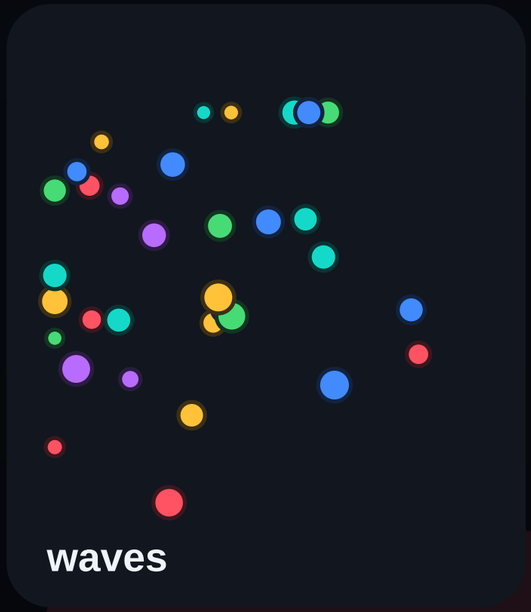
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